# SIEV-Net: A Structure-Information Enhanced Voxel Network for 3D Object Detection From LiDAR Point Clouds

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***Abstract*— As one of the fundamental tasks in scene understanding, 3D object detection from LiDAR point clouds has drawn extensive attention in the past few years. Although the existing voxel-based methods have achieved remarkable performance, how to effectively exploit geometric structure information of the point clouds to boost the detection performance remains to be explored. In this article, we propose a novel structure-information enhanced voxel network (SIEV-Net) for 3D object detection from LiDAR point clouds. The proposed SIEV-Net learns feature representations of 3D objects by jointly considering uneven spatial distribution and height information of the point clouds. Specifically, considering the uneven spatial distribution characteristics of point clouds, a hierarchical-voxel feature encoding module is proposed to effectively extract features of voxels in both sparse and dense regions. Besides, by utilizing the bird's eye view (BEV) map of point clouds, a height information complement module is designed to minimize the height information lost in the process of point feature aggregation in a voxel network. Experimental results on the widely used KITTI benchmark dataset have demonstrated the efficacy of the proposed SIEV-Net.**

***Index Terms*— 3D object detection, LiDAR point clouds, scene understanding, structure information.**

## I. Introduction

With the advancement of machine learning and computer vision, scene understanding based on deep learning [1]–[3] has attracted considerable attention in recent years. As one of the hot researches in scene understanding, 3D object detection from LiDAR point clouds is widely used in remote sensing and autonomous driving due to its ability to analyze the 3D scene and accurately perceive the category and location information of objects [4]. Although many image-based methods have been proposed, their performance is limited due to the lack of geometric structural information in images [5], [6]. Unlike RGB images, LiDAR point clouds capture the spatial distribution and surface features of 3D objects through a set of points. It provides rich geometric structural information for understanding the real 3D scene [7]. However, the points in point clouds are generally unordered, sparse, and unevenly distributed. Such characteristics of point clouds should be effectively exploited to boost 3D object detection performance.

Recently, many 3D object detection works from LiDAR point clouds have been proposed and made remarkable progress. According to the input form, these methods can be divided into two categories, namely, raw point cloud-based methods and regular point cloud-based methods. Among them, raw point cloud-based methods [11]–[13] directly input the sampled point clouds and utilize a point-wise extraction network to learn point features. Then, the detection head predicts the detection results based on the point features. In practice, obtaining effective point features in the point clouds is computationally intensive, which limits the practicality of the raw point cloud-based methods.

As for the regular point cloud-based methods, typical regular representations of point clouds are bird's eye view (BEV) and voxel. BEV-based methods first project the height values of points into a 2D BEV map, and then, traditional 2D convolutions are used to extract high-level features of the BEV map for predicting the detection results [19], [20]. Although height information is well preserved in the BEV-based methods, projecting the height values of points into a BEV map inevitably loses partial horizontal position information. Compared with the BEV map, voxels are generally obtained by dividing irregularly point clouds into 3D grids where the coordinate values and reflection intensity are effectively retained. Based on the obtained voxels, a voxel feature encoding (VFE) layer is usually utilized to extract the features of sampled points in each non-empty voxel, and the point features are aggregated to generate the voxel feature representation [21]–[23]. Generally, 3D convolutions are used to compress the voxel feature maps to the 2D feature maps and followed by 2D convolutions to learn high-level semantic feature representation for 3D object detection. Particularly, by dividing the point clouds into pillars, voxel-based

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methods [24], [25] have achieved efficient detection by avoiding the use of 3D CNNs.

Despite that the recent advance in voxel-based methods have achieved, there are still several challenges to be addressed. First, points in the LiDAR point clouds are generally unordered, sparse, and unevenly distributed. Voxel-based methods generally divide the irregular point clouds into regular voxels of the same size, and the density of points in a voxel is inversely proportional to the distance from the sensor. Thus, voxels representing near objects have dense points, yet voxels representing distant objects contain sparse points. During voxel feature extraction, a fixed number of points in non-empty voxels are sampled, and then, features of these points are learned and aggregated into a voxel feature. For single-size voxels, fewer sampled points in feature extraction may result in loss of information for the voxels with dense points, and more sampled points would result in the voxels with sparse points being filled with many zeros, which will adversely affect the feature extraction. Therefore, taking point density into consideration in the voxelization of point clouds will be beneficial for extracting effective features for detection. Second, some voxel methods divide point clouds into pillars for the sake of detection efficiency, while the loss of height information affects the detection performance. Specifically, the height of the pillar is generally larger than its horizontal length and width. When aggregating the point-wise features, more loss of height information is expected than that of horizontal information. To address the issues of the uneven spatial distribution of points and partial loss of height information in the existing voxel-based methods, this article proposes a novel structure-information enhanced voxel network (SIEV-Net) by explicitly and jointly considering spatial distribution and height information of points. In summary, the main contributions are summarized as follows.

1) An SIEV-Net for 3D object detection from LiDAR point clouds is proposed by fully exploiting the geometric structure information of point clouds.
2) Considering the uneven spatial distribution of point clouds, a hierarchical-voxel feature encoding (HVFE) module is presented to effectively extract feature representations of voxels in both sparse and dense regions.
3) To enhance the height information of voxel-wise features, a simple yet effective height information complement (HIC) module is proposed by utilizing the BEV map of point clouds.

The rest of this article is organized as follows. Section II reviews the related work. Section III presents the proposed 3D object detection method in detail. The experimental results are evaluated in Section IV. The ablation study is reported in Section V. Finally, this article is concluded in Section VI.

## II. Related Work

### A. Raw Point Cloud-Based Methods

Benefiting from the powerful representation of point-wise features in point cloud segmentation and classification [8]–[10], raw point cloud-based methods aim to extract point features directly from point set for 3D object detection. Shi *et al.* [11] proposed PointRCNN, which directly generates high-quality 3D proposals from raw points in the first stage subnetwork and refines 3D proposals in the second stage subnetwork. Yang *et al.* [12] designed spherical anchors to obtain high-recall proposals and aggregated the features of points in proposals by PointsPool for refining the 3D proposals. Qi *et al.* [13] proposed a 3D object detection network, VoteNet, which predicts 3D boxes of objects by jointly utilizing a deep point set network and hough voting. Xie *et al.* [14] proposed a point-based attentive cont-conv fusion module to fuse the features of point clouds and images. Shi and Rajkumar [15] proposed a graph convolutional detection network to encode the point clouds compactly with graph representation. Zhou *et al.* [16] developed a detection framework with the assistance of instance segmentation and designed a spatial embedding strategy to efficiently predict instance segmentation. Huang *et al.* [17] designed a point-wise fusion module to enhance point features with semantic image features. In [18], geometric and semantic information about objects in images was explored by the designed 2D votes in images and utilized to assist a voting-based 3D detection pipeline. Vora *et al.* [40] enhanced the semantic information of point features by appending the classification scores of the image semantic segmentation output to the point clouds.

### B. BEV-Based Methods

BEV-based methods generate a BEV map by dividing the points into multiple slices and projecting the height values of points to the 2D grid of the corresponding slice. Then, the detection results are obtained by utilizing traditional 2D convolutions to extract high-level features of the BEV map. Yang *et al.* [20] designed a real-time 3D detection method from the BEV map by integrating specific detection tasks into an existing 2D convolution network. In [26], 2D object detector YOLOv2 [27] was extended to a 3D detection task by introducing a specific Euler-region-proposal network to estimate the pose of the object. Yang *et al.* [28] proposed an efficient detection method, which extracts geometric and semantic features of point clouds by combining BEV representations and estimated high-definition maps from the point clouds. Besides, methods using BEV maps and RGB images have also been studied in the past few years. For instance, Chen *et al.* [19] proposed a multi-view 3D network (MV3D), which projects the point clouds to the BEV map and combines the features of the BEV maps and images to refine 3D proposals in the BEV branch. To further utilize information on BEV maps and RGB images, Ku *et al.* [29] took RGB images and BEV maps as inputs, and generated 3D proposals according to the fused multi-modality features. Liang *et al.* [30] explored the geometry of point clouds by mapping the point clouds into ground-relative BEV representation and completing dense depth information of the point clouds. Although the BEV-based methods mentioned above have driven the development of 3D object detection, their performance is limited since projecting points into a BEV map inevitably leads to some loss of horizontal position information of the points.

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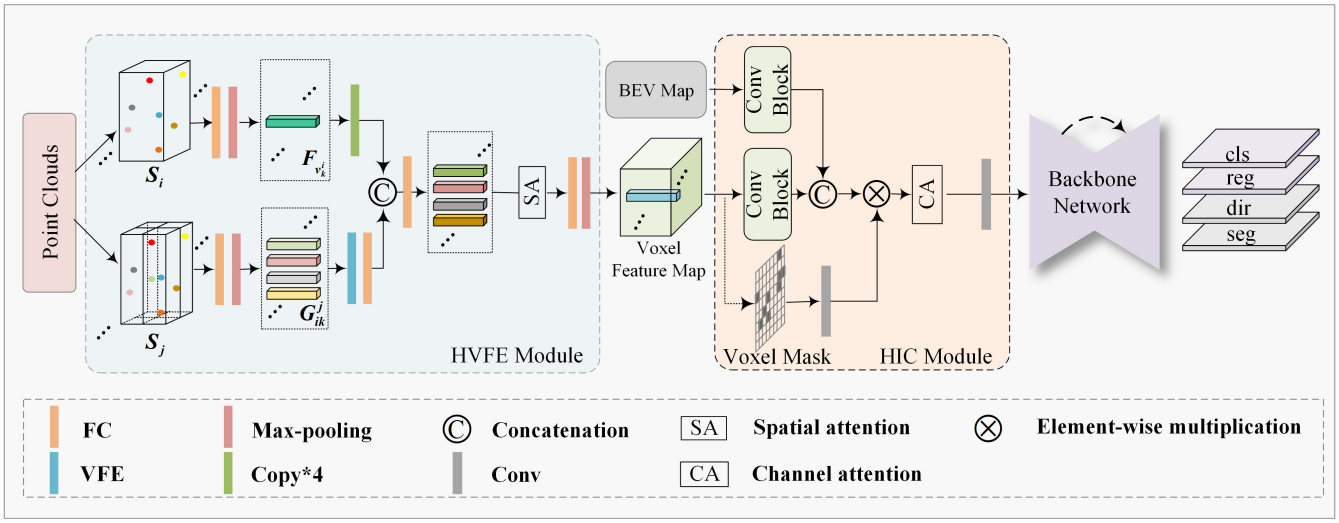



Fig. 1. Full pipeline of SIEV-Net. The main components of the network are the HVFE module, the HIC module, the 2D backbone, and the multi-task detection heads.

### C. Voxel-Based Methods

Recently, many voxel-based methods have been proposed, in which irregular points are divided into 3D grids. Zhou and Tuzel [21] proposed an end-to-end VoxelNet, which learns discriminative voxel features through a VFE layer and utilizes 3D convolutional layers to learn high-level features for detecting 3D objects. To reduce the complexity of VoxelNet, Yan *et al.* [22] designed sparse convolutional middle layers that are composed of sparse convolution and submanifold convolution [31], and realized real-time 3D object detection. Chen *et al.* [23] proposed a fast detection method based on PointRCNN by replacing a point-wise region proposal network with a voxel-wise region proposal network. Instead of using 3D convolution, some methods have been proposed to divide the point clouds into special voxels, such as pillars, so that only 2D convolution is required for efficient 3D object detection. For instance, Lang *et al.* [24] presented a detection method based on pillars, in which features of stacked pillars are learned and scattered back to a 2D pseudo-image feature map based on the indexes of pillars. After that, dense 2D convolution is used instead of 3D convolution. Besides, Wang *et al.* [32] designed a multi-scale voxel feature aggregation method, Voxel-FPN, which aggregates 2D feature maps of multi-scale voxels step by step and detects 3D objects in multi-resolution maps. Liu *et al.* [25] studied the robustness of 3D object detection from point clouds and designed a triple attention module to enhance the crucial information of voxels. Although voxel-based methods are effective in exploring geometric features in point clouds, these methods generally ignore the distribution characteristics of points, and height information of the point clouds is inevitably lost through pooling in the process of voxel-wise feature extraction.

## III. Proposed Method

### A. Architecture of SIEV-Net

The architecture of the proposed SIEV-Net is shown in Fig. 1. Given a LiDAR point cloud scene, the proposed SIEV-Net aims to predict 3D boxes of objects by an HVFE module and an HIC module to effectively exploit geometric information of the point clouds. In the HVFE module, the point clouds are first divided into hierarchical voxels with different scales. Then, voxel features at different scales are extracted separately, and voxel-wise attention is introduced to fuse the voxel features at different scales. The fused hierarchical-voxel features can be effectively extracted from unevenly distributed points for all non-empty voxels. The HIC module is to effectively exploit height information of the BEV map and horizontal structural information of the voxels. Specifically, the voxel feature map and the voxel mask are utilized to guide the learning of the height-enhanced voxel feature. By jointly utilizing the HVFE and HIC modules, the geometric structure information of the voxels in point clouds is expected to be adequately utilized. Subsequently, the backbone network with multi-task detection heads is employed to predict the 3D boxes of objects.

### B. Hierarchical-Voxel Feature Encoding Module

In voxel-based methods, effective extraction of voxel features is essential for obtaining robust detection results. However, the uneven distribution of points makes the extraction difficult and challenging. To deal with this, the HVFE module is proposed. In the HVFE module, the unevenly distributed point clouds are first divided into hierarchical voxel sets of two scales, namely, $S_i$ and $S_j$. Let $v_k^i \in R^{w_i \times l_i \times h_i}$ and $v_t^j \in R^{w_j \times l_j \times h_j}$ denote the $k$-th voxel at scale $S_i$ and the $t$-th voxel at scale $S_j$, respectively, where $w_i$ ($w_j$), $l_i$ ($l_j$), and $h_i$ ($h_j$) denote the width, length, and height of $v_k^i$ ($v_t^j$). For scales $S_i$ and $S_j$, the size of the hierarchical voxels are defined as $w_i = 2w_j$, $l_i = 2l_j$, and $h_i = h_j$. In point clouds, objects in dense point regions have rich information, and small-scale voxels are beneficial to learn fine feature representations of the objects. However, objects in sparse point regions contain relatively small amounts of information; thus, large-scale voxels are needed to capture the reliable

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information for detection. In particular, features of points in voxels are extracted, respectively, and then, the voxel features are obtained by aggregating the point-wise features in voxels of each scale. Let $p_{v_k^i}$ denote the point set in the non-empty voxel $v_k^i$, and the voxel feature $F_{v_k^i}$ is obtained by

$$F_{v_k^i} = \phi\left(\sigma\left(f\left(p_{v_k^i}\right)\right)\right) \tag{1}$$

where $f(.)$ denotes a fully connected (FC) layer used to learn the high-dimensional representation of points, $\sigma(.)$ denotes a nonlinear activator, and $\phi(.)$ denotes a max-pooling operation used to aggregate point features into voxel feature representation. Similarly, the voxel features at scale $S_j$ is obtained.

Since the voxel features at different scales effectively characterize voxels in the dense and sparse point regions, fusing the voxel features of different scales is helpful to enrich the information of the objects at different distances and positions in the scene. Considering the spatial hierarchical relationship between voxels at $S_i$ and $S_j$, voxel features at scale $S_j$ that correspond to the spatial position of non-empty voxel $v_k^i$ at scale $S_i$ are first grouped as $G_{ik}^j$. The grouped voxels have the same spatial size as $S_i$. Then, a VFE layer [21] is conducted on $G_{ik}^j$ to obtain the voxel features of different receptive fields, and an FC layer is followed to keep the same feature dimension as $S_i$. Finally, the processed $G_{ik}^j$ is concatenated with the feature $F_{v_k^i}$ along the channels, and a voxel-wise spatial attention is utilized to obtain the fused hierarchical-voxel features $F_k$, which are formulated as

$$F_k = \zeta\left(f\left[F_{v_k^i}, \psi\left(G_{ik}^j\right)\right]\right) \tag{2}$$

where $\psi(.)$ denotes the operation including a VFE layer and an FC layer, [.] denotes the concatenation, and $\zeta(.)$ is composed of a spatial attention [33], an FC layer, and a max-pooing layer to obtain the fused hierarchical-voxel features.

### C. Height Information Complement Module

By utilizing the proposed HVFE module, fused hierarchical-voxel features are obtained. However, the height of point clouds in each voxel is generally larger than that of horizontal width or length. This means that height information is less emphasized or lost partially when aggregating point features within the voxel. The HIC module is designed to compensate for such a loss.

Generally, the BEV map of point clouds is obtained by projecting the height values of points into a 2D map, thereby retaining the height information of the points. The HIC module utilizes this property of the BEV map. First, according to the index of the spatial position of the voxel in the point cloud data, fused hierarchical-voxel features $F_k$ are scattered to obtain a voxel feature map [24]. Then, the high-level feature $F_b$ of the BEV map and the high-level feature $F_v$ of the voxel feature map are extracted by using a ConvBlock, respectively, where $F_b$ better retains the height information and $F_v$ effectively captures the horizontal position information. $F_b$ and $F_v$ are concatenated along the channel dimension to form a fused semantic feature representation $F_S = [F_b, F_v]$. In this article, considering the sparsity of the point clouds, a mask map $M$ is defined to indicate whether the corresponding voxel is empty or not. Specifically, the mask map $M$ is composed of 0 and 1, where 1 indicates that the voxel is not empty. The masked fusion feature $F_m$ is given as follows:

$$F_m = F_S \otimes \tau(M) \tag{3}$$

where $\tau(.)$ represents a $3 \times 3$ convolution with stride 2 and an activation, and $\otimes$ denotes element-wise multiplication.

To capture geometric information from $F_m$, a channel attention is applied on $F_m$ to obtain effective feature $F_{E_S}$ as output of the HIC module

$$F_{E_S} = F_m + F_m \otimes C_a(F_m) \tag{4}$$

where $C_a(.)$ denotes the channel attention [33].

### D. Multi-task Detection Head

Following the HVFE and HIC modules, a backbone network with multi-task detection heads is utilized to detect 3D objects. The multi-task detection heads include classification, regression, direction angle classification, and voxel-wise segmentation.

The classification branch aims to generate the classification scores of the predicted boxes. Considering the imbalance of positive and negative samples in the classification, focal loss [34] is used as the classification loss $L_{\text{cls}}$. The regression branch aims to predict the center coordinates, length, width, height, and direction angle of the regression offset of the anchors. The regression labels are the offsets between the predefined positive anchor and the corresponding ground-truth box. Similar to the method in [22], smooth L1 loss is used to calculate the regression loss $L_{\text{reg}}$. The direction angle classification branch aims to determine whether the predicted boxes are flipped, and softmax loss [24] is utilized as the direction angle classification loss $L_{\text{dir}}$. In addition, an auxiliary task of voxel-wise segmentation is employed to assist the network to learn discriminative features by judging whether the voxel belongs to the foreground objects. Particularly, the voxel-wise segmentation is only applied to the non-empty voxels. For robust segmentation, we ignore the segmentation loss of the voxel near the ground-truth box via horizontally enlarging the ground-truth box by one voxel. Considering the imbalance between foreground and background voxels, the focal loss [34] is used as the segmentation loss $L_{\text{seg}}$.

Finally, jointly utilizing the multi-task detection heads, the total loss function of the proposed SIEV-Net is defined as

$$L_t = \frac{1}{N_p^a}\left(L_{\text{cls}} + \beta_1 L_{\text{reg}} + \beta_2 L_{\text{dir}}\right) + \frac{1}{N_f^v}\beta_3 L_{\text{seg}} \tag{5}$$

where $N_p^a$ denotes the number of positive anchors, $N_f^v$ denotes the number of foreground voxels, and $\beta_1$, $\beta_2$, and $\beta_3$ denote the parameters to balance the losses of different tasks.

### E. Implementation Details

In this article, the backbone network with a UNet-like architecture [35] is used to extract high-level semantic features

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of the point clouds. Specifically, two convolution blocks are utilized to extract the bottom-up multi-scale features, and the multi-scale features are progressively fused in a top-down manner. Each convolution block contains one convolutional layer with the kernel size of $3 \times 3$ and stride size of 2, and five convolutional layers with the kernel size of $3 \times 3$ and stride size of 1. Besides, the ConvBlock in the HIC module contains one convolutional layer with the kernel size of $3 \times 3$ and stride size of 2, and three convolutional layers with the kernel size of $3 \times 3$ and stride size of 1. Meanwhile, each convolution layer in the convolution block is followed by a batch normalization and an ReLU activation.

For the point clouds, each point has nine feature elements that are coordinates, reflectance, distance from the point to the arithmetic mean of all points in the voxel, and horizontal offsets from point to voxel center. Following [24] and [25], the height of voxels is set to be the same as that of the detection range. Besides, the division of the positive and negative anchors is based on the 2D Intersection over Union (IoU) [36], [37] between anchors and ground-truth boxes, and a single anchor box with two orientations (0° and 90°) is adopted in the experiments. For "Cars," the range of detection in the point clouds is set to [(0, 70.4), (−40, 40), (−3, 1)] m, the size of the anchor is set to (width = 1.6 m, length = 3.9 m, and height = 1.5 m), and the height of the anchor center is −1 m. The IoU thresholds for the positive and negative anchors are set as 0.6 and 0.45, and the other anchors are ignored. For scales $S_i$ and $S_j$, 100 and 35 points are sampled for the voxels, respectively. For "Pedestrians" and "Cyclists," the range of point clouds is set to [(0, 48), (−20, 20), (−2.5, 0.5)] m, and the IoU thresholds for positive and negative anchors are set as 0.5 and 0.35. For scales $S_i$ and $S_j$, 100 and 30 points are sampled for the voxels, respectively. In the experiments, the size of the anchors is set to (width = 0.6 m, length = 0.8 m, and height = 1.73 m) and (width = 0.6 m, length = 1.76 m, and height = 1.73 m) for "Pedestrians" and "Cyclists," respectively, and the heights of the anchor centers are both set to −0.6 m.

All experiments were performed using GeForce GTX 1080 Ti GPU and Intel i7-8700K processor @ 3.70 GHz. The optimizing process of the proposed SIEV-Net is implemented by adopting adaptive moment estimation (Adam) [38] with an initial learning rate of 0.0002. The proposed network is trained for 200 epochs with a batch size of 2. For every 15 epochs, the learning rate is decayed by a factor of 0.8.

## IV. Experiments

In this section, experimental settings, including dataset, evaluation metrics, and parameter selection, are presented. Then, the results are presented and compared with those obtained by state-of-the-art methods. Some detection results of the proposed method are also visualized. Subsequently, robustness studies are reported to show the stableness of the proposed method. Finally, the running times of different methods are presented to prove the efficiency of the proposed SIEV-Net.

### A. Experimental Settings

*1) Dataset:* In this article, the widely used KITTI Object Detection Benchmark dataset [39] is adopted to evaluate the proposed SIEV-Net. The dataset consists of 7481 training samples and 7518 test samples with three categories of objects, "Cars," "Pedestrians," and "Cyclists." According to the object size, occlusion state, and truncation level, three levels of difficulty, Easy, Moderate, and Hard, are defined for each category. Similar to the setup in [25], the training samples are divided into a training set with 3712 samples and a validation set with 3769 samples for preliminary verification of the proposed method. For data augmentation, global flipping, rotation, and scaling are applied to the point clouds. Specifically, ground-truth samples are randomly selected and placed into other point clouds to simulate crowded scenes. When evaluating the detection performance on the test set, the training samples are resplit into the training set and the validation set at the ratio of 5:1, which is the same as the setting in [25].

*2) Evaluation Metrics:* Following the official evaluation protocol of the KITTI dataset, 3D average precision (AP) is utilized to evaluate the detection performance of the different methods [39]. The IoU thresholds of the AP metric are set as 0.7 for "Cars" and 0.5 for "Pedestrians" and "Cyclists." Since AP calculated with 40 recall positions has been the official evaluation metric, the detection performances of different methods on the test set are evaluated by the 3D AP calculated with 40 recall positions. Besides, for a fair comparison with existing methods, the detection performance on the validation set is calculated with 11 recall positions. In addition, the mean values of AP (mAP) at three difficulty levels are also computed to illustrate the overall performance.

*3) Parameter Selection:* In the total loss function [see (5)], there are three weights, $\beta_1$, $\beta_2$, and $\beta_3$, for regression loss, direction loss, and segmentation loss, respectively. To find suitable $\beta_1$, $\beta_2$, and $\beta_3$ for balancing the contributions of the different loss terms, greedy search experiments are conducted using the "Cars" category on the validation set. Specifically, $\beta_2$ and $\beta_3$ are first set to 0 to obtain an optimal $\beta_1$. As shown in Fig. 2, the best detection performance is obtained when $\beta_1$ is set to 2.0. Then, $\beta_1$ is fixed at 2.0, and the optimal $\beta_2$ is determined as 0.2. Finally, fixing $\beta_1$ and $\beta_2$, $\beta_3$ is found to be the best at 1.0. Thus, $\beta_1$, $\beta_2$, and $\beta_3$ are set to 2.0, 0.2, and 1.0 in the rest experiments.

### B. Comparison Results

In this section, extensive experiments are reported to show the effectiveness of the proposed method. Comparison methods include MV3D [19], F-PointNet [41], AVOD-FPN [29], PI-RCNN [14], UberATG-MMF [30], PointPainting [40], Point-GNN [15], SERCNN [16], VoxelNet [21], SECOND [22], PointPillars [24], and TANet [25].

Table I shows the detection performances of the proposed method and comparison methods on the validation set. As can be seen from the table, by only using LiDAR point clouds, the proposed SIEV-Net obtains 81.18%, 66.04%, and 71.44% 3D mAP on "Cars," "Pedestrians," and "Cyclists," respectively. Compared with the typical voxel-based method,

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TABLE I
COMPARISON OF 3D OBJECT DETECTION ON "CARS," "PEDESTRIANS," AND "CYCLISTS" OF THE VALIDATION SET (%). L AND R IN THE TABLE REPRESENT LIDAR POINT CLOUDS AND RGB IMAGES, RESPECTIVELY

| Method | Type | Data | Cars | | | | Pedestrians | | | | Cyclists | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Easy | Mod. | Hard | 3D mAP | Easy | Mod. | Hard | 3D mAP | Easy | Mod. | Hard | 3D mAP |
| F-PointNet [41] | Point | L&R | 83.76 | 70.92 | 63.65 | 72.78 | - | - | - | - | - | - | - | - |
| Point-GNN [15] | Point | L | 87.89 | 78.34 | 77.38 | 81.20 | - | - | - | - | - | - | - | - |
| MV3D [19] | BEV | L&R | 71.29 | 62.68 | 56.56 | 63.51 | - | - | - | - | - | - | - | - |
| AVOD-FPN [29] | BEV | L&R | 84.41 | 74.44 | 68.65 | 75.83 | - | - | - | - | - | - | - | - |
| VoxelNet [21] | Voxel | L | 81.98 | 65.46 | 62.85 | 70.10 | - | - | - | - | - | - | - | - |
| SECOND [22] | Voxel | L | 88.06 | 77.64 | 75.25 | 80.32 | 61.20 | 54.14 | 51.37 | 55.57 | 81.12 | 60.31 | 58.86 | 66.76 |
| PointPillars [24] | Voxel | L | 87.50 | 77.01 | 74.77 | 79.76 | 66.73 | 61.06 | 56.50 | 61.43 | 83.65 | 63.40 | 59.71 | 68.92 |
| TANet [25] | Voxel | L | 88.21 | 77.85 | 75.62 | 80.56 | **70.80** | 63.45 | 58.22 | 64.16 | **85.98** | 64.95 | 60.40 | 70.44 |
| SIEV-Net | Voxel | L | **88.91** | **78.25** | **76.39** | **81.18** | 70.43 | **66.52** | **61.16** | **66.04** | 85.90 | **66.48** | **61.95** | **71.44** |

TABLE II
RESULTS OF 3D OBJECT DETECTION ON "CARS," "PEDESTRIANS," AND "CYCLISTS" OF THE TEST SET (%). L AND R IN THE TABLE REPRESENT LIDAR POINT CLOUDS AND RGB IMAGES, RESPECTIVELY

| Method | Type | Data | Cars | | | | Pedestrians | | | | Cyclists | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Easy | Mod. | Hard | 3D mAP | Easy | Mod. | Hard | 3D mAP | Easy | Mod. | Hard | 3D mAP |
| F-PointNet [41] | Point | L&R | 82.19 | 69.79 | 60.59 | 70.86 | 50.53 | 42.15 | 38.08 | 43.59 | 72.27 | 56.12 | 49.01 | 59.13 |
| PI-RCNN [14] | Point | L&R | 84.37 | 74.82 | 70.03 | 76.41 | - | - | - | - | - | - | - | - |
| PointPainting [40] | Point | L&R | 82.11 | 71.70 | 67.08 | 73.63 | 50.32 | 40.97 | 37.87 | 43.05 | 77.63 | 63.78 | 55.89 | 65.77 |
| Point-GNN [15] | Point | L | 88.33 | 79.47 | 72.29 | 80.03 | 51.92 | 43.77 | 40.14 | 45.28 | 78.60 | 63.48 | 57.08 | 66.39 |
| SERCNN [16] | Point | L | 87.74 | 78.96 | 74.30 | 80.33 | - | - | - | - | - | - | - | - |
| MV3D [19] | BEV | L&R | 74.97 | 63.63 | 54.00 | 64.20 | - | - | - | - | - | - | - | - |
| AVOD-FPN [29] | BEV | L&R | 83.07 | 71.76 | 65.73 | 73.52 | 50.46 | 42.27 | 39.04 | 43.92 | 63.76 | 50.55 | 44.93 | 53.08 |
| UberATG-MMF [30] | BEV | L&R | 88.40 | 77.43 | 70.22 | 78.68 | - | - | - | - | - | - | - | - |
| VoxelNet [21] | Voxel | L | 77.82 | 64.17 | 57.51 | 66.50 | - | - | - | - | - | - | - | - |
| SECOND [22] | Voxel | L | 84.92 | 73.47 | 68.72 | 75.70 | 44.68 | 36.98 | 34.71 | 38.79 | 75.25 | 54.45 | 48.70 | 59.47 |
| PointPillars [24] | Voxel | L | 82.58 | 74.31 | 68.99 | 75.29 | 51.45 | 41.92 | 38.89 | 40.09 | 77.10 | 58.65 | 51.92 | 62.56 |
| TANet [25] | Voxel | L | 84.39 | 75.94 | 68.82 | 76.38 | 53.72 | 44.34 | 40.49 | 46.18 | 75.70 | 59.44 | **52.53** | 62.56 |
| SIEV-Net | Voxel | L | **85.21** | **76.18** | **70.60** | **77.33** | **54.00** | **44.80** | **41.11** | **46.64** | **78.75** | **59.99** | 52.37 | **63.70** |

i.e., VoxelNet [21], SECOND [22], PointPillars [24], and TANet [25], the proposed SIEV-Net achieves 11.08%, 0.86%, 1.42%, and 0.62% improvements in 3D mAP on "Cars" category, respectively. The main reason is that the proposed HVFE module has considered the point cloud distribution so that the effective voxel features from the point clouds can be extracted. Besides, the HIC module in the proposed SIEV-Net effectively utilizes the height information of the object, which is beneficial to effectively perceive and detect the 3D objects in the point clouds. Compared with the recent TANet [25], the proposed SIEV-Net obtains better performances on the "Hard" subset of the relatively small objects, "Pedestrians" and "Cyclists." Overall, by considering the non-uniform distribution of the point clouds and the impact of the aggregation operation in feature extraction, the proposed SIEV-Net achieves competitive performance on the KITTI validation set. Fig. 3 visually shows some examples of predicted 3D boxes of the proposed SIEV-Net and the ground-truth 3D boxes on the validation set. As can be seen in the figure, the proposed method achieves good results for hard and challenging objects. For instance, the first line gives the successful detection of a heavily occluded car, in which the points are truncated. The proposed SIEV-Net effectively detects the occluded object, which further verifies the effectiveness of the proposed method.

Table II shows the comparison results of the test set. It can be observed that the proposed SIEV-Net achieves better 3D mAP compared with the other voxel-based methods, such as VoxelNet [21], SECOND [22], PointPillars [24], and TANet [25]. Compared to PointPillars [24] and TANet [25], the proposed SIEV-Net achieves 2.04% and 0.95% increases in 3D mAP on "Cars," 6.55% and 0.46% increases in 3D mAP on "Pedestrians," and 1.14% and 1.14% increases in 3D mAP on "Cyclists." Compared with the raw point cloud-based method, Point-GNN [15], the proposed SIEV-Net achieves competitive detection performance on "Pedestrians." Compared with the multi-modality detection methods, UberATG-MMF [30] and PointPainting [40], the 3D mAP of SIEV-Net is slightly lower on "Cars" and "Cyclists," respectively.

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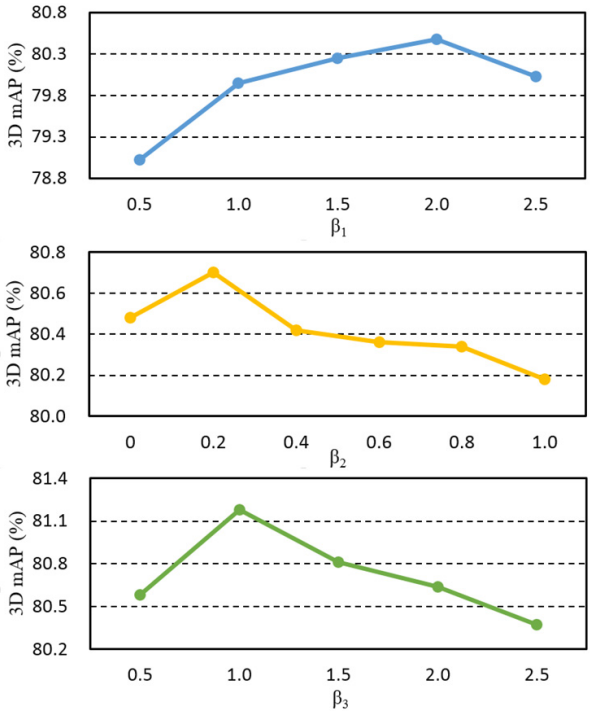



Fig. 2. Parameter analysis of the proposed method.

TABLE III
ROBUSTNESS STUDY OF THE PROPOSED SIEV-NET WITH DIFFERENT NOISE POINTS (%)

| Method | Number of noise points | Easy | Mod. | Hard | 3D mAP |
|---|---|---|---|---|---|
| TANet | 50 | 87.3 | 77.36 | 75.54 | 80.07 |
| SIEV-Net | | **88.07** | **77.96** | **75.63** | **80.55** |
| TANet | 100 | 86.89 | 77.21 | 75.47 | 79.86 |
| SIEV-Net | | **87.54** | **77.87** | **75.57** | **80.33** |
| TANet | 150 | 86.5 | 77.13 | 75.26 | 79.63 |
| SIEV-Net | | **86.71** | **77.78** | **75.55** | **80.01** |

The reason may be that UberATG-MMF and PointPainting have leveraged additional information from RGB images. Fig. 4 shows some examples of predicted 3D detection results on the test set. Since the ground-truth boxes of the 3D objects in the test set are not provided, only the predicted boxes are shown. It can be observed that the proposed SIEV-Net can effectively detect the objects.

### C. Analysis of Robustness

In this section, the robustness of the proposed SIEV-Net is evaluated using the "Cars" category on the validation set by adding random noise points around each object in the point cloud scene. The way to add noise points is the same as that used in TANet [25], which generates uniformly distributed random noise points around the 3D objects. As can be seen from Table III, the detection performance of both TANet and SIEV-Net decreases slightly with the increasing number of noise points. However, the proposed SIEV-Net maintains its better performance than TANet in all cases.

TABLE IV
COMPARISON OF AVERAGE GPU TIME

| Method | GPU | Frames/s |
|---|---|---|
| Point-GNN [15] | GTX 1080Ti | 2 |
| SECOND [22] | GTX 1080Ti | 20 |
| PointPillars [24] | GTX 1080Ti | 42 |
| TANet [25] | GTX 1080Ti | 27 |
| SIEV-Net | GTX 1080Ti | 22 |

TABLE V
EVALUATION OF THE KEY COMPONENTS OF THE SIEV-NET ON THE "CARS" CATEGORY (%)

| Methods | Easy | Mod. | Hard | 3D mAP |
|---|---|---|---|---|
| w/o HVFE and w/o HIC | 88.02 | 77.23 | 70.76 | 78.67 |
| w/HVFE | 88.34 | 77.77 | 75.52 | 80.54 |
| w/HIC | 88.09 | 77.82 | 75.95 | 80.62 |
| SIEV-Net | **88.91** | **78.25** | **76.39** | **81.18** |

### D. Analysis of Computational Complexity

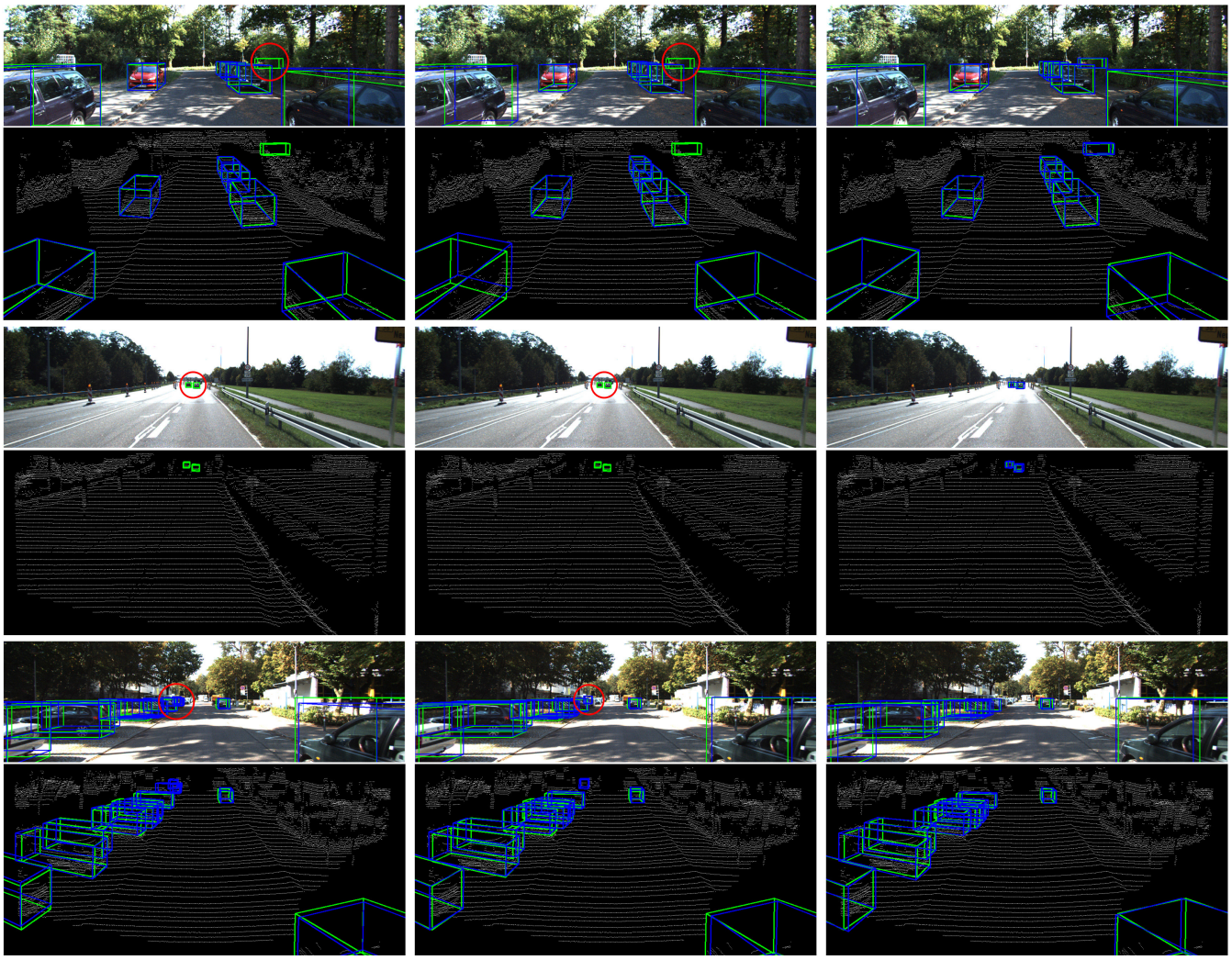
In practice, detection speed is another important indicator to measure the ability of a detection method. In this section, to fairly compare the detection speed, we obtain the inference times of the compared methods by running the official codes under the same platform. It can be observed from Table IV that the proposed SIEV-Net achieves 22 frames/s. Compared with the raw point cloud-based method, Point-GNN [15], the proposed SIEV-Net has a remarkable advantage in inference time. Compared with PointPillars [24], the proposed SIEV-Net is mainly time-consuming in the HVFE module. In the HVFE module, the features of multi-scale voxels are encoded and fused, thus taking more running time. Compared with SECOND [22] and TANet [25], the proposed SIEV-Net achieves better 3D object detection performance with the same level of computational complexity.

## V. ABLATION STUDY

### A. Evaluation of Key Components

In order to illustrate the effectiveness of the key components in the proposed SIEV-Net, ablation studies are conducted on the validation set using the "Cars" category. Results are presented in Table V. Specifically, the baseline method refers to the SIEV-Net without the HVFE module and HIC module, and the voxel features are extracted without utilizing the hierarchical division of voxels and height information contained in the BEV map. Besides, "w/HVFE" and "w/HIC" refer to the methods including the HVFE module and HIC module in the process of the voxel feature extraction, respectively. As shown in Table V, both the HVFE and HIC modules contribute to the performance improvement, and the proposed SIEV-Net with both the HVFE and HIC modules obtains

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(a) (b) (c)

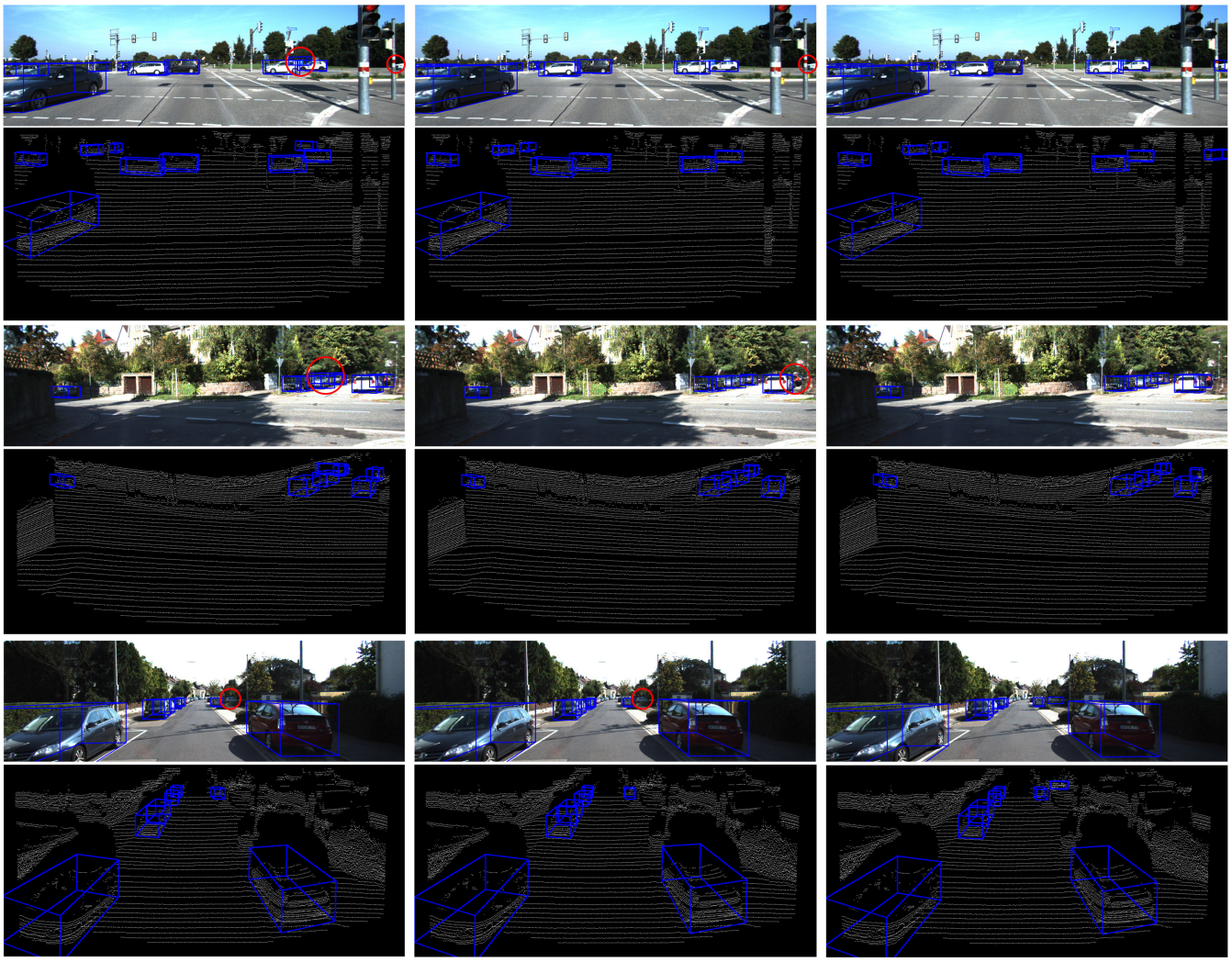
Fig. 3. Visualization of detection results on the validation set. For each scene, the point cloud and the corresponding RGB image are visualized. (a) Results of PointPillars. (b) Results of TANet. (c) Results of the proposed SIEV-Net. The ground-truth 3D boxes and the predicted 3D boxes are drawn with green and blue bounding boxes, respectively, and some failure cases of the compared methods are shown with red circles.

the best detection performance. Compared with the baseline method, the method "w/HVFE" achieves a 1.87% improvement in 3D mAP. Especially for the "Hard" subset, "w/HVFE" achieves a 4.76% improvement in 3D mAP. This is mainly because effective voxel features are extracted from unevenly distributed points to describe 3D objects in both sparse and dense point regions. In addition, the method "w/HIC" achieves 0.07%, 0.59%, and 5.19% improvements in 3D AP on "Easy," "Mod.," and "Hard" subsets over the baseline, respectively. It implies that the HIC module effectively enhances the feature representations of 3D objects by compensating the height information for the voxel feature map. Importantly, with both the HVFE and HIC modules, the proposed SIEV-Net achieves 0.89%, 1.02%, and 5.63% improvements in 3D AP on "Easy," "Mod.," and "Hard" subsets, respectively, and obtains an overall performance improvement of 2.51% in 3D mAP. As expected, the largest improvement is on the hard cases. This demonstrates that the two components proposed in the SIEV-Net are indeed beneficial for perceiving the geometric structure of 3D objects, thereby contributing to the performance improvement of 3D object detection.

### *B. Evaluation of Multi-Scale Setting*

To verify the impact of the multi-scale setting in the HVFE module, the experiments of the proposed HVFE module taking the multi-scale voxels of different sizes as inputs are conducted using the "Cars" category on the validation set. Results are shown in Table VI, in which S refers to the horizontal sizes of one scale voxel as $\{0.16, 0.16\}$ $m^2$, "S + 1/4S" denotes that the sizes of multi-scale voxels are 1 times S and 1/4 times S, and so on. As shown in Table VI, when there are two scale voxels in the HVFE module, adding a 1/2S scale voxel to the S scale voxel can obtain the best detection performance. When the sizes are set too large or too small, the detection performance will decrease. Comparing the case of "S + 1/2S," the cases of "S + 1/4S" and "S + 2S" decrease by 0.35% and 1.32% in 3D mAP, respectively.

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(a) (b) (c)

Fig. 4. Visualization of detection results on the test set. For each scene, the point cloud and the corresponding RGB image are visualized. (a) Results of PointPillars. (b) Results of TANet. (c) Results of the proposed SIEV-Net. The predicted 3D boxes are drawn with blue bounding boxes. For intuitive observation, some failure cases of the compared methods are shown with red circles.

TABLE VI
ANALYSIS OF THE SIZE OF MULTI-SCALE SETTING (%)

| Voxel scales | Easy | Mod. | Hard | 3D mAP |
|---|---|---|---|---|
| S+1/4S | 88.42 | 77.92 | 76.14 | 80.83 |
| S+1/2S | **88.91** | **78.25** | **76.39** | **81.18** |
| S+2S | 87.18 | 77.53 | 74.88 | 79.86 |
| 2S+1/2S | 86.23 | 75.62 | 74.19 | 78.68 |

TABLE VII
ANALYSIS OF THE NUMBER OF MULTIPLE SCALES (%)

| Voxel scales | Easy | Mod. | Hard | 3D mAP |
|---|---|---|---|---|
| 1/2S | 88.24 | 77.48 | 75.40 | 80.37 |
| S | 88.09 | 77.82 | 75.95 | 80.62 |
| S+1/2S | **88.91** | **78.25** | **76.39** | **81.18** |
| S+1/2S+1/4S | 88.53 | 77.91 | 76.29 | 80.91 |

In order to evaluate the impact of the number of multiple scales in the HVFE module, related experiments are conducted, and the results are shown in Table VII. As shown in the table, the performance of multi-scale voxels is generally higher than that of a single-scale setting. This is mainly because multi-scale voxels facilitate learning more effective voxel features in both sparse and dense regions. In addition, the detection performance with three-scale voxels "S + 1/2S + 1/4S" is lower than that of the two-scale voxels "S + 1/2S." This is probably because the voxel feature extraction becomes more complex as the number of voxels increases. Therefore, increasing the number of voxels is likely to bring difficulties to the network learning with the HVFE module.

## VI. CONCLUSION

In this work, a novel SIEV-Net is proposed for 3D object detection from LiDAR point clouds, in which geometric structure information of the point clouds is effectively exploited to learn effective feature representations of 3D objects. Specifically, an HVFE module is proposed to extract voxel features

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from an unevenly distributed point cloud so that the features of the voxels in both sparse and dense regions can be effectively extracted. Besides, considering the height information loss caused by point feature aggregation, an HIC module is designed to enhance the height information in the hierarchical-voxel features. Experimental results on the widely used benchmark dataset have demonstrated the efficacy of the proposed SIEV-Net.

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